RelayAI Router Explorer

Installation guide — run the router explorer on your own laptop, macOS or Windows.

What this is

A small web app that shows the trained LLM router making one decision at a time: it embeds your question, scores every model in the pool, adds the cost term, and calls the argmin. It runs entirely on your machine. Two pieces start separately — a **Python backend** that holds the trained network, and a **web frontend** you open in a browser.

Before you start

- **Python 3.10 or newer** and **Node.js 18 or newer**. Verified on Python 3.13 and Node 20.
- **Internet on the first run only.** The backend downloads the sentence encoder (`all-MiniLM-L6-v2`, about 87 MB) once. Every run after that works offline.
- **About 1.5 GB of disk**, most of it PyTorch. No GPU is needed; everything is CPU only.
- You need **two terminal windows** open at the same time, one per piece.

Step 0 — unpack the zip

Unzip `RelayAI-Router-Explorer.zip`. Everything below assumes the folder ended up in your **Downloads**. If you put it elsewhere, change the path in the first `cd` command and keep the rest identical.

Check what you already have

Open a terminal and run these. If both print a version number, skip to the install.

MACOS Terminal (press `Cmd + Space`, type *Terminal*)

```
python3 --version
node --version
```

WINDOWS PowerShell (press `Win`, type *PowerShell*)

```
python --version
node --version
```

> **Missing one?** Install Python from python.org/downloads and Node from nodejs.org (take the LTS build). On Windows, tick **"Add python.exe to PATH"** in the Python installer, then close and reopen PowerShell. On macOS with Homebrew you can instead run `brew install python node`.

MACOS Running the app

Terminal 1 — the backend

1. **Go to the folder.**

```
cd ~/Downloads/RelayAI-Router-Explorer
```

2. **Create a private Python environment**, so these packages do not touch the rest of your system.

```
python3 -m venv .venv
source .venv/bin/activate
```

   Your prompt now starts with `(.venv)`.

3. **Install the Python packages.** A few minutes the first time — PyTorch is large.

```
pip install --upgrade pip
pip install -r app/backend/requirements.txt
```

4. **Start the backend.**

```
python -m uvicorn app.backend.main:app --port 8000
```

   Wait for `Application startup complete`. On the very first run it downloads the encoder before that line appears. **Leave this window running.**

Terminal 2 — the frontend

Open a **second** Terminal window (`Cmd + N`). Do not close the first one.

1. **Go to the frontend folder.**

```
cd ~/Downloads/RelayAI-Router-Explorer/app/frontend
```

2. **Install the web packages** (about 120 of them, some seconds).

```
npm install
```

3. **Start it.**

```
npm run dev
```

4. **Open the app** at http://localhost:5173.

WINDOWS Running the app

PowerShell 1 — the backend

1. **Go to the folder.**

```
cd $HOME\Downloads\RelayAI-Router-Explorer
```

2. **Create a private Python environment.**

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

If that is blocked with a message about running scripts being disabled, allow it for this window only, then run the activate line again:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

3. **Install the Python packages.** A few minutes the first time.

```
python -m pip install --upgrade pip
pip install -r app\backend\requirements.txt
```

4. **Start the backend.**

```
python -m uvicorn app.backend.main:app --port 8000
```

Wait for `Application startup complete`. **Leave this window running.**

PowerShell 2 — the frontend

Open a **second** PowerShell window. Do not close the first one.

1. **Go to the frontend folder.**

```
cd $HOME\Downloads\RelayAI-Router-Explorer\app\frontend
```

2. **Install the web packages.**

```
npm install
```

3. **Start it.**

```
npm run dev
```

4. **Open the app** at http://localhost:5173.

Check the backend on its own

In a third window, this should print a block of JSON naming the encoder and the pool size.

MACOS

```
curl http://127.0.0.1:8000/health
```

WINDOWS

```
curl.exe http://127.0.0.1:8000/health
```

Stopping, and starting again later

Press `Ctrl + C` in each of the two windows. On macOS you can then type `deactivate` to leave the Python environment.

Next time you do not repeat the installs. Only these:

MACOS

```
# terminal 1
cd ~/Downloads/RelayAI-Router-Explorer
source .venv/bin/activate
python -m uvicorn app.backend.main:app --port 8000

# terminal 2
cd ~/Downloads/RelayAI-Router-Explorer/app/frontend
npm run dev
```

WINDOWS

```
# powershell 1
cd $HOME\Downloads\RelayAI-Router-Explorer
.\.venv\Scripts\Activate.ps1
python -m uvicorn app.backend.main:app --port 8000

# powershell 2
cd $HOME\Downloads\RelayAI-Router-
Explorer\app\frontend
npm run dev
```

If something goes wrong

WHAT YOU SEE	WHAT TO DO
A red banner: *cannot reach the router backend at http://127.0.0.1:8000*	The backend is not running, or it is still downloading the encoder. Go back to terminal 1 and wait for `Application startup complete`. Start the backend before the frontend.
`address already in use` on port 8000	Something else holds that port. Use another one, and tell the frontend where to look. MACOS `python -m uvicorn app.backend.main:app --port 8010`, then in terminal 2 `VITE_API_BASE=http://127.0.0.1:8010 npm run dev` WINDOWS same first line, then `$env:VITE_API_BASE="http://127.0.0.1:8010"; npm run dev`
Port 5173 already in use	Vite picks the next free port by itself and prints it. Open the address it prints, not 5173.
`python: command not found` or `node: command not found`	Not installed, or the terminal was open before you installed it. Close every terminal, open a new one, try again. On macOS use `python3`, on Windows `python`.
running scripts is disabled on this system (Windows)	Run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` in that window, then activate again. It applies only to that window.
The pip install seems stuck	PyTorch is several hundred megabytes. Give it a few minutes on a normal connection before assuming it has failed.
A warning about `HF_TOKEN` / unauthenticated requests	Harmless. It is the encoder download noting you are not signed in to Hugging Face. Nothing to do.
The page loads but every panel is empty	The frontend started before the backend was ready. Reload the browser once the backend says `Application startup complete`.

What is in the folder

PATH	WHAT IT IS
`app/backend/`	The HTTP shell. `main.py` serves the router, `requirements.txt` lists the Python packages.
`app/frontend/`	The web interface: the three panels you see in the browser.
`relay_router.py`	The router itself. Every number in the app comes from here — no scoring happens in the browser.
`router_bundle/`	The trained network and the measured tables it reads. Already fitted; nothing trains when you run the app.